AI BOOTCAMP CAPSTONE · 2026

SAP Basis
L1 Support Agent

Autonomous diagnosis + human-gated remediation for SAP infrastructure incidents

LangGraph | OpenAI (configurable) | Qdrant | Langfuse | SSH / sapcontrol

01 THE PROBLEM

L1 Support is slow, repetitive, and high-stakes

Mean Time to Diagnose

SAP Basis engineers spend 30-60 min per incident gathering data that could be collected in seconds by automated tooling.

Tribal Knowledge

Root cause patterns (known dumps, disk pressure cycles, hung WPs) live in engineers' heads or in scattered runbooks. Not where the LLM can reach them.

Risk of Unreviewed Fixes

A single wrong sapcontrol command can bring down a system. Automatic L1 fixes without a review step are a liability.

For whom: SAP Basis engineers and L1 support teams managing SAP systems on Linux.

02 THE SOLUTION

Multi-agent diagnosis + gated remediation

TRIGGER
CLI / Webhook

- SSH / sapcontrol
- Filesystem checks
- Log reader

SUPERVISOR
Agent

ROOT CAUSE ANALYSIS
Agent

- Qdrant RAG
- Runbooks define fixes

HUMAN
Approval Gate

- LangGraph interrupt()
- Operator sees full RCA before any action

REMEDIATION
Agent

- Runbook fix_command
- Executed via real SSH
- Verify after fix

REPORT
Output

Why LLM?

Log analysis and root cause synthesis require natural language reasoning — connecting a SIGSEGV in dev_w2 to a known SAP Note is not a regex problem. A plain script can collect data; it takes an LLM to interpret it and explain the cause to a human operator.

📊 Langfuse — full trace per incident: span per node, token cost, eval scores, operator approve/reject logged as feedback

What you'll see in the terminal

demo — bash

```
$ python run_agent.py --host p23cs.example.com --sid P23 --alert 'WP02 not responding'
  Mode: PRODUCTION (real SSH + RAG + LLM)

[SUPERVISOR] Incident received. Routing to RCA...
[RCA] ssh_run: sapcontrol -nr 00 -function GetProcessList
[RCA] ssh_run: sapcontrol -nr 00 -function ABAPGetWPTable
[RCA] read_file: /usr/sap/DEV/work/dev_w2
[RCA] rag_lookup: 3 results for "WP02 not responding"
[RCA] Root cause: SIGSEGV in Z_CUSTOM_FM

[SUPERVISOR] Routing to human_approval

⚠  APPROVAL REQUIRED
   Proposed Fix: restart_workprocess
   Fix Command : sapcontrol -nr 00 -function
                 RestartService
   Risk        : MEDIUM

Approve? [yes/no]: _
```

① Trigger

--scenario shortcut or explicit --host/--alert

② RCA Agent

SSH to Docker mock host + Qdrant RAG

③ Approval Gate

Graph pauses via LangGraph interrupt()
Operator sees full evidence before deciding

④ After 'yes'

Executes fix_command from runbook via SSH
then runs verify_command to confirm

Where can this go wrong?

🔍 RAG quality depends on ingestion quality

If runbooks are sparse or SAP Notes aren't loaded, retrieval will miss matches. The LLM will still generate an answer. However, it won't be grounded in your specific environment. Mitigation: evaluation pipeline that flags low-confidence retrievals.

🤖 LLM hallucination on unfamiliar dumps

Novel dump patterns not in training data or Qdrant will get plausible-sounding but potentially wrong root cause hypotheses. The human approval gate is the last line of defense, and it must never be bypassed.

🔐 SSH credential scope

The agent needs OS-level SSH access. In production this must be a locked-down service account with exactly the permissions needed.
A compromised agent with broad SSH rights is a serious risk.

📡 No RFC / ABAP layer

Scope is intentionally limited to OS + sapcontrol. SM50/SM66 work process details, ABAP lock entries, and spool administration require RFC. This agent cannot see or fix those.

Privacy & Regulatory: SAP system logs can contain personal data (user IDs, transaction data). GDPR applies if running in EU. All data should stay on-prem — no log content sent to external APIs without data classification review.

04 WHAT SUCCESS LOOKS LIKE

Metrics, eval, and next steps

< 60s
Time-to-diagnosis
vs 30–60 min manual

> 80%
RCA confidence
scored per trace in Langfuse

100%
Human approval
before any write action

Evaluation Pipeline (Langfuse)

RCA confidence
Scored high/medium/low per trace — track trend over time

RAG relevance
Best Qdrant similarity score logged as Langfuse score

Operator approval rate
Approve/reject logged per trace — high rejection = wrong fixes

MTTD
Seconds from alert to RCA complete — scored per trace

Next Steps if taken further:

RFC/BAPI layer · Slack/Teams alert integration · Multi-SID fleet support · Fine-tuned SAP embeddings · ✅ Hybrid mode · ✅ 9 runbooks + RAG chunking · ✅ Runbook-driven fixes · ✅ Docker mock SAP host

Thank You

Questions?

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